

NeKoRAG Technical Design Document

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1. Architecture Overview

NeKoRAG adopts a three-stage retrieval pipeline. Stage 1 performs dual-path recall: dense semantic search via ChromaDB and sparse keyword search via BM25. Stage 2 merges results using Reciprocal Rank Fusion (k=60). Stage 3 applies cross-encoder reranking followed by context expansion along linked-list chunk pointers.

2. Document Ingestion Flow

1. Upload document (.md/.txt/.pdf) via REST API
2. Parse content with format-specific loader
3. Chunk text using Markdown-header-aware splitter
4. Generate dense embeddings via Ollama + bge-m3
5. Store vectors in ChromaDB (persistent mode)
6. Build BM25 index with jieba tokenization

3. Evaluation Results

3.1 Per-Category MRR Comparison

Category	Dense	BM25	Hybrid	Reranker
Keyword (n=4)	0.833	0.875	0.875	1.000
Semantic (n=4)	0.875	0.646	0.875	0.800
Mixed (n=4)	1.000	1.000	1.000	1.000
Overall (n=12)	0.903	0.840	0.917	0.933

3.2 Key Findings

The evaluation demonstrates clear complementary behavior between dense and sparse retrieval. On keyword-precise queries, BM25 achieves a higher MRR (0.875) than pure Dense (0.833). On semantic queries, Dense significantly outperforms BM25 (0.875 vs 0.646). Hybrid retrieval with RRF maintains the best of both, and cross-encoder reranking further improves overall MRR to 0.933 -- a 3.3% improvement over pure Dense and 11.1% over pure BM25.